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Hybrid intelligent water Drops algorithm for examination timetabling problem

Bashar A. Aldeeb^{a,*}, Mohammed Azmi Al-Betar^{b,c}, Norita Md Norwawi^d, Khalid A. Alissa^{a,e}, Mutasem K. Alsmadi^f, Ayman A. Hazaymeh^g, Malek Alzaqebah^{h,i}

^a Deanship of Information and Communication Technology, Imam Abdulrahman Bin Faisal University, Dammam, Saudi Arabia

^b Artificial Intelligence Research Center (AIRC), College of Engineering and Information Technology, Ajman University, Ajman, United Arab Emirates

^c Department of Information Technology, Al-Huson University College, Al-Balqa Applied University, Al-Huson, Irbid, Jordan

^d Universiti Sains Islam Malaysia, Faculty of Science and Technology, Sembilan, Malaysia

^e College of Computer Science and Information Technology, Imam Abdulrahman Bin Faisal University, Dammam, Saudi Arabia

^f Department of Management Information Systems, College of Applied Studies and Community Service, Imam Abdulrahman Bin Faisal University, Saudi Arabia

^g Department of Mathematics, Faculty of Science and Information Technology, Jadara University, Jordan

^h Department of Mathematics, College of Science, Imam Abdulrahman Bin Faisal University, P.O. Box 1982, Dammam, Saudi Arabia

ⁱ Basic and Applied Scientific Research Center, Imam Abdulrahman Bin Faisal University, P.O. Box 1982, Dammam, Saudi Arabia

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ABSTRACT

The current study investigates the Intelligent Water Drops (IWD) metaheuristic algorithm to construct and produce good quality solutions for the university examination timetabling problem (UETP). The IWD is a population-based metaheuristic that simulates the dynamic of the river systems. The main motivations for investigating IWD algorithm for examination timetabling problem is the ability to explore the search space effectively. The main drawback of IWD algorithm is like other population-based algorithm in exploitation process where it is very efficient scanning several search space niches, but it is unable to drilling down in each niche to which it navigates. In this paper we propose a hybrid approach based on IWD and locale search algorithm to improve the exploitation of IWD algorithm. The experimental results demonstrated that the proposed algorithm (i.e., Hybrid IWD) obtained best results in three datasets when comparing with the best-known results performed by the swarm intelligent approaches. Finally, the proposed algorithm achieved one best results in comparison with the other metaheuristic approaches.

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1. Introduction

Examination timetabling considered to be complex and time-consuming task required by every educational organization each semester (Mandal et al., 2020; Qu et al., 2009b), which defined as “the scheduling for the exams of a set of university courses, avoiding overlap of exams of courses having common students, and spreading the exams for the students as much as possible” (Schaefer, 1999). Examination timetabling problem is a complex problem to be tackled

led due to huge count of students. The same problem also occurs among moderate-sized universities (Mandal et al., 2020). On the other hand, the manual timetable requires a big extent of effort to complete within limited days. The scheduling process of examination that the faculty is concerned with, in some universities, is only scheduled for the specialized courses while the university students are scheduled for the common courses. Hence, it becomes the reason for having an unsuitable timetable for the students as the closed dated examinations are scheduled by this manual scheduling. It should be noted that the administrator wants the timetable to be shortened (June et al., 2019b).

When the timetable is centrally prepared, students could face many problems in their preparations. For instance, students may find complications in the timetable when they want to make possible change as per their requirements. The phenomenon of complexity is reflected by these situations of the scheduling of the examination timetable which may fulfil the people's preferences (Battistutta et al., 2017). The exam timetabling, in computing

* Corresponding author.

E-mail addresses: Baaldeeb@iau.edu.sa (B.A. Aldeeb), m.albetar@ajman.ac.ae (M. Azmi Al-Betar), Norita@usim.edu.my (N. Md Norwawi).

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terms, is assumed to be NP-hard optimization problems. It has a deep search space and hence, it requires a good mechanism with a strong search ability to produce a sufficient solution (Mandal et al., 2020).

Moreover, to solve the examination timetabling problems, large numbers of mechanisms are implemented in the literature review (Al-Betar, 2020; Alzaqebah & Abdullah, 2015; Bellio et al., 2021; Eng et al., 2020; Leite et al., 2018; Mandal et al., 2020; Qu & Burke, 2009; Tilahun, 2019). The technique that provides the motivation for using a good local search neighborhood structure for IWD algorithm in search space of examination timetabling that seems deep and huge (Qu & Burke, 2009). In this approach, the researchers empower the exploitation capacity by modifying the algorithm. As demonstrated by the recent studies, the hybridization between the single and population-based approaches reveals that it may generate competitive performance in obtaining the expected solutions (Blum & Roli, 2003). The motivation is surpassed by hybridizing the proposed IWD to descend with local search method to optimum solution quickly.

This study tried to improve the performance of IWD algorithm to efficiently tackle examination timetabling problem (UETP). In order to improve the exploitation capability for IWD algorithm that proposed in (Bashar A. Aldeeb et al., 2015) which is the only adapted version of IWD was used to solve UETP, this study hybridized the local search optimizer (LSO) and the IWD algorithm to tackle UETP. The aim of this hybridization is to strike the right balance between exploration and exploitation. Thus, to quickly discover the search space and to get better exploitation. Therefore, the premature convergence can be avoided.

2. Related works

The related work of the main topics provided in this paper is given. These topics are Examination timetabling problem, IWD and hybrid IWD. Their related works are summarized in the following subsections.

2.1. University examination techniques

Examination timetabling is the process of assigning exams to timeslots (and rooms) in accordance with hard constraints related to the schedule violations and soft constraints related to the schedule quality. Examination timetabling problem can be uncapacitated or capacitated due to the examination's room capacity consideration (Lei et al., 2018). The room capacities are ignored in the uncapacitated problem version, while the capacitated examination timetabling problem has to include the hard constraints in a manner that the count of pupils in a particular room that they are allocated to, should not during the period as per scheduled exceed the room capacity (Leite et al., 2019; Pillay & Banzhaf, 2009). However, only the Uncapacitated University Examination Timetabling Problem (UETP) is considered in our study.

A various number of approximation techniques for the UETP have been established by the Artificial Intelligence (AI) and the Operational Research (OR) communities. A recently proposed survey is available in (Aldeeb et al., 2019). Prior establishment using the graph coloring heuristics, which assigns exams one by one to timeslots, based on the level of complexity. As a recovery approach to the timetable with unscheduled exams, a backtracking method is often used with these techniques. Carter and Laporte (1996) initiated the main research on the UETP by incorporating several graph coloring heuristic methods to UETP (Carter et al., 1996). Other studies incorporating the graph coloring heuristic methods for UETP are also involved (Asmuni et al., 2009; Bellio et al., 2021).

2.2. The previous applications of IWD algorithm

The natural phenomenon i.e. Water drops swarm through the soil into the river bottom, attracted the researcher's attention to be simulated by the IWD algorithm with the soil onto the riverbed (Hosseini, 2007). It was investigated that employing the IWD concept showed tremendous success in numerous discrete optimization problems (Alijla et al., 2013; Duan et al., 2009; Hosseini, 2007; Niu et al., 2012; Shah-Hosseini, 2010) and machine learning tasks (Shah-Hosseini, 2012b). This leads to its application in the continuous problems of optimization (Shah-Hosseini, 2012a). This is because of the advantage of IWD compared to other optimization methods (Shah-Hosseini, 2010, 2012b). It incorporates a less complex although wholesome mathematical model. Also, it is easily to implement and applicable to many discrete and continuous optimization problems and it instantly converge them into the optimal solution. In addition, It works in the construction of the solution in the population based on the performed data based on the familiarity search iteration instead of taking into account the improvement of the current population (Shah-Hosseini, 2013).

The IWD algorithm is performed successfully to a number of optimization problem such as Economic Load Dispatch (ELD) problem, Traveling Salesman Problem (TSP), Feature Selection (FS), Maximum Clique Problem (MCP), Web services composition problem, Stainer tree problem, Single Ucarv smooth path planning problem, Vehicle Routing Problem (VRP), Multilevel Thresholding problems, K-means Clustering Problem, Mobile Ad-hoc networks (MANET). However, based on its sophistication nature of an optimization problems, the method has also been hybridized and altered to increase quality of the solutions. However, it also needs to be noted that the behavior of IWD still require more theoretical analysis (Hosseini, 2007).

2.3. Hybrid metaheuristics for UETP

The process of hybridizing a population-based technique with local search-based technique has drawn more attention (Blum & Roli, 2003). This hybridization is designed to striking a balance in the processes of exploitation and exploration to gain the benefits of local search-based and population-based technique. In general, many trends in research are highlighting the efficacy of the local search-based techniques with the population-based techniques. For instance, Blum and Roli (2003) wrote, "*population-based techniques are better in identifying promising areas in the search space, whereas trajectory methods are better in exploring promising areas in the search space. Thus, metaheuristic hybrids that in some way manage to combine the advantage of population-based techniques with the strength of trajectory techniques are often very successful*".

Population-based techniques can explore many regions of the search space at the same time. Though, they are poor at discovering a local optimal solution for each search space region (Fesanghary et al., 2009). Conversely, local search-based techniques can fine-tune the region of search to which they converge fast and produces the local optimal solution. Nevertheless, local search-based techniques search over paths in the search region without performing a the full search of the whole search region (Abualigah & Diabat, 2021; Al-Betar et al., 2012; June et al., 2019a).

Hybridization of IWD Algorithm with sequential quadratic programming as a local optimizer was firstly introduced by Nofereesti and Shah-Hosseini (2012) who expanded the classical IWD algorithm for Steiner Tree Problem (STP) to overcome some shortcomings in the convergence behavior. At each iteration, this hybridization influence on the convergence and the performance of the original IWD algorithm. As results show that the algorithm with faster convergence speed has better performance than the traditional heuristics and other local search algorithms. In addition,

Mokhtari (2015) hybridized the basic IWD algorithm with an Iterated Local Search (ILS) algorithm to solve a generalized type of order scheduling problem where denial of received orders is permitted with a penalty. At the production period start, a group of orders are determined by the manufacturer. Because of the capacity constraints, a subset of orders only can be processed by the manufacturer and the manufacturer must choose to refuse some of unwanted orders. Accepted orders are proceeded to be scheduled by a set of parallel identical processors. The goal is to choose the best orders set with high contribution in the benefit of the manufacturer and then find the suitable schedule of accepted orders diminishing the number of tardy orders. The experimental results show the effectiveness of hybridized of IWD with ILS.

3. UETP description and formulation

UETP refers to the assignment of examinations to a limited number of timeslots in such a way that there is no clash between examinations. No conflict refers to the situation where no student is needed to take more than one examination in the same timeslot. The standard benchmark for UETP was introduced by Carter et al. (1996) and will be described in Section 5.

The notations for UETP formulation are shown in Table 1. A timetable solution is represented as a vector $\mathbf{x} = (x_1, x_2, \dots, x_E)$, where x_i is the timeslot assigned to exam i , $i = \{1, 2, \dots, E\}$. The satisfaction of hard constraint in the solution of the UETP is given by the following equation.

$$\forall x_i \in \mathbf{x} \wedge C_{ij} \geq 1 \quad x_i f = \forall x_i$$

The problem can be formulated as the minimization of the sum of proximity costs as formulated in the following equation (Carter et al., 1996).

$$\min f(\mathbf{x}) = \frac{\sum_{i=1}^{E-1} \sum_{j=i+1}^E C_{ij} \times \text{Prox}_{ij}}{N} \quad (1)$$

The value of the proximity cost function $f(\mathbf{x})$ is referred as the Penalty Value (PV) of a feasible timetable. The previous equation represents the examination cost, which is the proximity value, multiplied by the number of conflicting students.

4. Proposed Hybrid IWD algorithm (Hybrid IWD) for UETP

Despite of the ability of global search and converging rate of the IWD algorithm that used before in solving UETP (Aldeeb, 2016), it still getting some trapped in the local optimum because it has weakened to exploit the search space areas containing the better solutions. There-

Table 1

The notations used in the UETP formulations according to Carter et al. (1996).

Symbols	Definition
T	The total number of timeslots.
N	The total number of students.
E	The total number of exams.
t	Set of timeslots $P = \{1, 2, \dots, T\}$.
S	Set of students $S = \{1, 2, \dots, N\}$.
e	Set of exams $e = \{1, 2, \dots, E\}$.
$\mathbf{x}$	A timetable solution is given by $\mathbf{x} = (x_1, x_2, \dots, x_E)$.
x_i	The timeslot of exam i where $i = \{1, 2, \dots, E\}$.
prox_{ij}	Proximity coefficient matrix element: whether the timetable $\mathbf{x}$ is penalized based on the distance between time-period of exam i in time-period of exam j .
	where: $\text{Prox}_{ij} = \begin{cases} 2^{5- x_i-x_j }, & \text{if } 1 \leq x_i-x_j \leq 5 \\ 0, & \text{Otherwise} \end{cases}$
u_{ij}	Student exam matrix element: whether student S_i is sitting for exam j .
	$u_{ij} = \begin{cases} 1, & \text{if } x_i = x_j \\ 0, & \text{Otherwise} \end{cases}$
C_{ij}	Conflict matrix element: total number of students sharing exam i and exam j . $C_{ij} = \sum_{k=1}^N u_{ki} \times u_{kj} \forall i, j \in E$

fore, Hybrid IWD with local search algorithm is most probably achieved the right balance between exploration and exploitation. Here, the researchers will initially be discussed how and where the local search optimizer can be hybridized with the IWD algorithm (the proposed algorithm will be called Hybrid IWD algorithm). Thereafter, the process of local search optimizer is thoroughly describe including the most effective neighborhood structure used.

4.1. IWD for uncapacitated examination timetabling problem

The original IWD algorithm is customized to meet the characteristics and requirements of UETP. The main customization is done by maintaining feasibility during the IWD search using saturation degree. The saturation degree is embedded into the customized IWD algorithm.

The Adapted IWD has six main phases, each phase consists of several steps. The following phases and steps explain how the Adapted IWD algorithm is customized to be workable for UETP.

Phase 1: Static Parameters Initialization. Which are the unchanged parameters during the search using IWD algorithm such as:

- it_{max} : maximum number of iterations which is normally set by the user.
- $Drops$: the number of intelligent water drops $(x_1, x_2, \dots, x_{Drops})$ which denote to the number of timetabling solutions.
- a_{val}, b_{val} and c_{val} : velocity updating parameters which are a set of parameters used to control the velocity update function.
- a_{soil}, b_{soil} and c_{soil} : soil updating parameters which are a set of parameters used to control the soil update function.
- $InitialSoil$: which is the initial value of the local soil, such that the soil of the path between every two exams i and n_e that is set by $IWDsoil(i, n_e) = InitialSoil$.
- $\rho IWD_s, s \in \{1, 2, \dots, Drops\}$: the global soil updating parameter, which is chosen from the range of $[0, 1]$.

Algorithm 1: The main steps for the proposed Adapted IWD algorithm.

1. Initialize static parameters.
2. for $(NI = 1 \text{ to } iter_{max})$ do
3. Initialize dynamic parameters.
4. Spread one random timeslot (Tslot) to a random exam for all IWD_s .
5. Update the list of visited node $V_s(IWD)$.
6. for $(e = 1 \text{ to } M - 1)$ do
7. for $(s = 1 \text{ to } Drops)$ do
8. Determine next exam n_e to be scheduled in the solution IWD_s using saturation degree principle.
9. update Velocity of the solution IWD_s .
10. update Soil of the solution IWD_s .
11. end for
12. end for
13. select the local best solution in the iteration population T^{lb} .
14. update the soil value of all edges included in the T^{lb} .
15. update the global best solution T^{Tb} .
16. if value of $T^{Tb} > T^{lb}$ then
17. $T^{Tb} = T^{lb}$
18. end if
19. end if
20. Stops with the global best solution.

Phase 2: Dynamic parameters initialization. the parameters are initialized at the beginning of solution construction and are updated during the construction process. They are reverted to their initial values at the beginning of each iteration. The dynamic parameters are:

- $V_s(IWD)$: A list of exams that are already chosen by the solution IWD_s , where $s \in \{1, 2, \dots, Drops\}$.
- $InitialVel$: The initial velocity of solution IWD_s .
- $IWDsoil^{IWD}$: The initial soil loaded to solution IWD_s .
- T^{lb} : Local best solution, which is reinitialized after each iteration of the IWD algorithm. T^{lb} represents the least penalty value of the timetable solutions reached in such iterations.
- T^{lb} : Global best solution is the best timetabling solution in the population which has the minimum penalty, which is also reinitialized after each iteration of the IWD algorithm.

Phase 3: Spread one random timeslot to a random exam for all IWD_s . In this phase Adapted IWD assigns a random timeslot to a random exam to be the first visited exam for each IWD_s (See algorithm 2), where $s \in \{1, 2, \dots, Drops\}$. Furthermore, the list $V_s(IWD)$ is updated to include the visited exam for each solution s .

Algorithm 2: Proposed Pseudo-code for spreading random timeslot to random exam.

1. $s = 0$
2. While $s < Drops$ do
3. rnd exam = select random exam from the list of exams.
4. rnd Tslot = selected random Tslot from the list of timeslots.
5. Assign the Tslot rnd Tslot to $IWD_s(rndexam)$.
6. Update the list of visited exams $V_s(IWD)$.
7. Increment the counter s .
8. End while

Phase 4: Solutions construction: The main aim of this phase is to construct the IWD solutions. Note that, the visited-exams array $V_s(IWD)$ is updated. Then, at each step of the construction phase, the timetabling solution extends the partial solution by traversing a new exam, (i.e., a feasible component that does not violate any hard constraints of the problem). As shown in algorithm 1 lines (6 – 12), the construction process includes two loops, where in the outer loop, each exam will be assigned by a timeslot, and in the inner loop, each solution will be constructed exam by exam using the saturation degree (SD). The construction step is completed by the transition of all timetable solutions through the graph until the stopping criteria for constructing a complete population is met. The construction phase consists of the following steps:

Determine exam n_e to be scheduled in the solution IWD_s using SD The major adaption of the Adapted IWD algorithm is done in this step by determining an exam n_e to be scheduled in the solution IWD_s using the SD principle.

In SD, the matrix $IWD_Exam_Tslot(Drops, M, P)$ is used to maintain the feasibility through the construction process. The $IWD_Exam_Tslot(Drops, M, P)$ is binary matrix where:

$$IWD_Exam_Tslot(s, e, p) = \begin{cases} 1, & \text{If exam can be feasibly scheduled in timeslot } p \\ 0, & \text{Otherwise} \end{cases} \quad (1)$$

Where $s \in \{1, 2, \dots, Drops\}$, $e \in \{1, 2, \dots, M\}$ and $p \in \{1, 2, \dots, P\}$. Note that, all elements of $IWD_Exam_Tslot(s, e, p)$ is initialized by 1 and they will be updated according to the conflict matrix during the solutions construction process, $IWD_Exam_Tslot(s, e, p)$ is very essential in the proposed algorithm because it can help in determining the next exam n_e that will be scheduled. The SD use $IWD_Exam_Tslot(s, e, p)$ to determine the exam n_e that has the least number of feasible timeslots using the following equation:

$$n_e(s) = \arg_{e \in [1, M]} \sum_{p=1}^P IWD_Exam_Tslot(s, e, p) \quad (2)$$

$p \in \{1, 2, \dots, Drops\}$

Where, $n_e(s)$ is the exam n_e in the solution IWD_s that has the least feasible timeslot which is denoted as the next exam. Furthermore, the matrix $IWD_Exam_Tslot(s, e, p)$ is used to assign a feasible timeslot τ to exam n_e directly by checking if the value of the exam $IWD_Exam_Tslot(s, e, p)$ is equal to (1), where $P \in \{1, 2, \dots, Drops\}$. Then, this timeslot is suitable to scheduling.

However, in case where there is more than one exam which has the same number of minimum timeslots, the proposed algorithm will select one of them according to the probability function using the following equations:

$$\rho IWD_s(n_e) = \frac{f(IWDsoil(i, n_e))}{\sum_{i \notin V_s(IWD)} f(IWDsoil(i, p))} \quad (3)$$

So that:

$$f(IWDsoil(i, n_e)) = \frac{1}{\mathcal{E}_s + g(IWDsoil(i, n_e))} \quad (4)$$

And

$$g(IWDsoil(i, n_e)) = \begin{cases} IWDsoil(i, n_e) & \text{if } \min_{l \in V_s(IWD)} (IWDsoil(i, l)) \geq 0 \\ IWDsoil(i, n_e) - \min_{l \notin V_s(IWD)} (IWDsoil(i, l)) & \text{Otherwise} \end{cases} \quad (5)$$

Exam n_e is a member of a vector $L \in \{L_1, L_2, \dots, L_k\}$ where k is the total number of exams that have the same number of the least feasible timeslots which will be added to the visited-exam array $V_s(IWD)$, after choosing the next exam n_e , the proposed algorithm will assign the exam n_e by feasible timeslot τ , which means that, the exam n_e of the least number of timeslots in the solution IWD_s will be assigned by feasible timeslot τ as in algorithm 3:

Algorithm 3: Proposed Pseudo-code for selecting random timeslot τ to be schedule for exam n_e .

1. **for** ($t = 0$ to P)
2. **If** $IWD_Exam_Tslot(s, n_e, t) = 1$
3. Push Tslot t into vector ϕ .
4. **endif**
5. **endfor**
6. $\tau = \text{select random element from } \phi$.

Not that, the $IWD_Exam_Tslot(s, e, p)$ will be updated according to the assigned exam n_e and the conflict matrix. For example, if the exam 3 in IWD_s is assigned by timeslot 2, the other exams that conflict with the exam n_e (i.e., $\eta \in \{e_1, e_2, \dots, e_r\}$) will update the corresponding timeslot 3 by the value 0 as follows:

$$\sum_{i=1}^r IWD_Exam_Tslot(5, i, 1) = 0e_i \in \eta \quad (6)$$

According to the process of solution construction that builds the solutions IWD_s exam by exam, if the feasibility of the IWD_s cannot be achieved, the reconstruction process will be activated to fix the feasibility. The velocity $IWD_s(velocity^{IWD_s}(t+1))$ is updated at every moving from exam i to exam n_e by:

$$velocity^{IWD_s}(t+1) = velocity^{IWD_s}(t) + \frac{a_{vel}}{b_{vel} + c_{vel} \times IWDsoil^2(i, n_e)} \quad (7)$$

where $velocity^{IWD_s}(t+1)$ is the updated velocity of the timetable solution IWD_s and $a_{vel} = 1$ and $b_{vel} = 0.01$ are the static parameters used to represent the nonlinear relationship between the velocity of solution s , and the inverse of the amount of soil in the local path, i.e. $IWDsoil(i, n_e)$. For the timetable solution s , when the Adapted IWD algorithm constructs the next exam n_e , both $IWDsoil^{IWD_s}$ and $IWDsoil(i, n_e)$ are updated by:

$$IWDsoil(i, n_e) = (1 - \rho) \times IWDsoil(i, n_e) - \rho \times \Delta IWDsoil(i, n_e) \quad (8)$$

$$IWDsoil^{IWD_s} = IWDsoil^{IWD_s} - \Delta IWDsoil(i, n_e) \quad (9)$$

where ρ is a small positive constant between zero and one, $\Delta IWDsoil(i, n_e)$ is the amount of soil removed from the local path and carried by a solution s . Note that, $\Delta IWDsoil(i, n_e)$ is nonlinearly proportional to the inverse of $velocity^{IWD_s}(t+1)$

$$\Delta IWDsoil(i, n_e) = \frac{a_{soil}}{b_{soil} + c_{soil} \times time(i, n_e; velocity^{IWD_s}(t+1))} \quad (10)$$

where, a_{soil} , b_{soil} , c_{soil} are the static parameters used to represent the nonlinear relationship between $\Delta IWDsoil(i, n_e)$ and the inverse of $velocity^{IWD_s}(t+1)$. Note that, $time(i, n_e; velocity^{IWD_s}(t+1))$ refers to the time needed for the timetable solution s to transit from exam i to exam n_e at $timeslot(t+1)$. It is defined as follows:

$$time(i, n_e; velocity^{IWD_s}(t+1)) = \frac{HUD(n_e)}{velocity^{IWD_s}(t+1)} \quad (11)$$

where $HUD(n_e)$ is a heuristic desirability degree of the edge between exam i and n_e . In timetabling we used $HUD(n_e)$ as a $f(IWD_s)$ after the next exam n_e is assigned. Note that, the $HUD(n_e)$ of IWD_s is a heuristic calculate the penalty value of the IWD_s partially constructed.

The processes of selecting exams to be visited as well as updating the velocity and local soil are iterated subject to the stopping criteria for obtaining a complete solution

Phase 5: Solutions enhancement: After the IWD_s are constructed in phase four, the solution enhancement phase will be started. As shown in algorithm 4 (Lines 13–19), the fittest solution of each IWD_s is determined as the local best solution, and is denoted as T^{lb} determined by the following equation:

$$T^{lb} = \arg \min_{s \in \{1, 2, \dots, Drops\}} (f(IWD_s)) \quad (12)$$

Where, $f(IWD_s)$ is the objective function which is used to evaluate the quality of the timetable solutions, and IWD_s is the timetabling solutions. To reinforce water drops in the subsequent iterations to follow T^{lb} and achieve the fittest solution over the iterations, the soil of all edges (the path between the current exam i and the next exam n_e) in T^{lb} is updated using the following equation. This update process is known as the global soil update.

$$IWDsoil(i, n_e) = (1 + \rho IWD_s) \times IWDsoil(i, n_e) - \rho IWD_s \times \frac{1}{E_{T^{lb}} - 1} \times IWDsoil_{T^{lb}} IWD_s \quad \forall (i, j) \in T^{lb} \quad (13)$$

where ρIWD_s is a positive constant and in each iteration, the best solution (global best), i.e., T^{lb} is either replaced by T^{lb} or maintained, as in the following equation:

$$T^{Tb} = \begin{cases} T^{lb}, & \text{If } f(T^{lb}) \leq f(T^{Tb}) \\ T^{Tb}, & \text{Otherwise} \end{cases} \quad (14)$$

Where the global best solution T^{Tb} is the solution that has the lowest penalty on all iterations while the local best solution T^{lb} is the best solution in each iteration. This means that we will find the global best solution T^{Tb} from all the solutions T^{IWD_s} by comparing the value of the local best solution T^{lb} with the value of the global best solution T^{Tb} . Then, the Adapted IWD algorithm will assign the value of the local best solution T^{lb} to the T^{Tb} if it was less than the old value for the global best solution T^{Tb} .

Phase 6: Termination: The solutions construction and enhancement phases, shown in algorithm 1 lines (6–19), are iterated until a termination condition is met.

The previously mentioned stages are produced Adapted IWD algorithm. At the same time, the researchers made an improvement on these steps to produce Modified IWD algorithm by using the concept of Global Best to diversity the solutions within the possible range of available values. This allows the Modified IWD algorithm to move towards the promising region of the solution search space and thus, helps to escape the local optimal, for more details about Modified IWD algorithm see Bashar A. AlDeeb (2016).

4.2. Local search optimizer (LSO)

In the Hybrid IWD algorithm, the major modification that applies to the Modified IWD algorithm was the hybridization of the Local Search Optimizer (LSO) with the Modified IWD algorithm. The position where the LSO is placed within the pseudo-code of the Hybrid IWD algorithm is after the solutions' construction phase as shown in Algorithm 4 lines (24–26). This combination aimed to balance between the exploration and the exploitation of the search space, which will enhance the solution quality.

Algorithm 4: The main steps for the proposed Hybrid IWD algorithm

```

1. Initialize static parameters.
2. for ( $NI = 1$  to  $Iter_{max}$ ) do
3. Initialize dynamic parameters.
4. Spread one random timeslot (Tslot) to a random exam
   for all  $IWD_s$ .
5. Update the list of visited node  $V_s(IWD)$ .
6. for ( $e = 1$  to  $E - 1$ ) do
7. for ( $s = 1$  to  $Drops$ ) do
8. Determine next exam  $n_e$  to be scheduled in the solution
    $IWD_s$  using saturation degree.
9. If  $IWD\_Exam\_Tslot(s, n_e, T^{Tb}(n_e)) = 1$  then
10.  $IWD_s(n_e) = T^{Tb}(n_e)$ 
11. Else
12. for ( $t = 0$  to  $P$ )
13. If  $IWD\_Exam\_Tslot(s, n_e, p) = 1$  then
14. Push Tslot  $t$  into vector  $\phi$ .
15. Endif
16. Endfor
17.  $\tau = \text{select random timeslot from } \phi$ 
18.  $IWD_s(n_e) = \tau$ 
19. Endif
20. update Velocity of the solution  $IWD_s$  (Eq. 7)

21. update Soil of the solution  $IWD_s$  (Eq. 8,9)
22. Endfor
23. Endfor (Eq. 1,2,3,4,5,6,7,8,9,10)
24. If  $RAND(0,1) < LSR$ 
25.  $IWD_s = LSO()$ 
26. Endif
27. update the soil value of all edges included in the  $T^{lb}$  (Eq. 12)
28. update the global best solution  $T^{Tb}$  (Eq. 13)
29. if value of  $T^{Tb} > T^{lb}$  then
30.  $T^{Tb} = T^{lb}$ 
31. Endif (Eq. 14)
32. Endfor
33. Stops with the global best solution.

```

Algorithm 4 shows how local search optimizer is hybridized with the Modified IWD algorithm as a new operator. In the Hybrid IWD algorithm, the local search optimizer is invoked to improve the constructed solutions locally with a probability rate of the Local Search Rate (LSR), where the LSR belongs to (0–1). Note that, the usage of the LSR parameter is to determine the utilization of the LSO. The percent of the LSR is equal to the probability of calling the LSO and thus, if the percentage of LSR was high, then the exploitation will be high. Whenever the LSO is called, the current solution is improved until the max iteration used for the LSO is reached.

The local search optimizer used the neighborhood structure, which is adapted in this version of the IWD to produce new set of neighboring solutions (See Algorithm 5). The detailed descriptions of the neighborhood structure and the process of generating neighboring solution are given in the next subsection. Local search or neighborhood search is a highly efficient metaheuristic framework to solve a large amount of optimization problems and constraint satisfaction (Hoos & Stützle, 2004). Local search algorithms try to progressively improve the working solution by searching for the neighbor's solutions around the working solution.

Consequently, the working solution is iteratively replaced by a neighbor solution, if it feasible and has better fitness value.

One of local search advantages is therefore the description of neighborhood method. Generally, a good neighborhood method provides effective search and lead accordingly to excellent results mostly is based on the initial solution. The weak neighborhoods search performance is very related to the initial solution (Papadimitriou & Steiglitz, 1998). Furthermore, the local search behavior strongly related to its neighborhood features. For example, some neighborhoods allow obtaining improvements of solution in an important and quick manner, nevertheless the improvement only happens for a restricted iterations number. In contrast, other neighborhoods allow only small improvements, but for a long time.

Algorithm 5: Proposed Local Search Optimizer (LSO) algorithm.

```

1. INPUT  $IWDs, f(IWDs)$ 
2. DO
3.  $i = \text{random integer number in range } [0 - 2]$ 
4. IF  $i = 0$  THEN
5.  $IWDs = LSO - SingleMove(IWD's)$ 
6. ELSEIF  $i = 1$  THEN
7.  $IWDs = LSO - Swap(IWD's)$ 
8. ELSE
9.  $IWDs = LSO - Kempe - Chain(IWD's)$ 
10. ENDIF
11. IF  $f(IWD's) > f(IWDs)$  THEN
12.  $IWDs = IWD's$ 
13.  $f(IWDs) = f(IWD's)$ 
14. ENDIF
15. WHILE (The  $Iter_{max}$  number is not met)

```

In UETP Neighborhood structure is utilized to get a new set of neighboring solutions through performing a small perturbation to a given solution and every neighborhood solution is found from a given solution by some kind of movement process (Laguna & Glover, 1993). Furthermore, if two or more neighborhoods provide complementary features, it is possible and interesting then to generate more powerful combined neighborhoods (Lü et al., 2011). So that, we investigate three existing neighborhoods in this study (Move, Swap, Kempe chain) inside the LSO (See Algorithm 5). LSO utilized these three neighborhood structures to discover the search region of the current drop thoroughly.

4.3. Hybrid IWD for UETP

Simply, the Local Search Optimizer (LSO) is hybridized in the improvement loop of adapted IWD as new phase (phase 6) to improve the produced timetabling solutions iteratively. By means of this hybridization mechanism, the exploitation capability of adapted IWD is empowered. After the solutions enhancement phase is completed, the LSO is invoked with a probability of Local Search Ratio (LSR) where $LSR \in [0,1]$ to improve the quality of each timetabling solution locally until the local optima is reached. The value of LSR determines the usage percentage of LSO. The larger the value of LSR, the heavier the usage of LSO.

5. Experiments and results

The performance of hybrid IWD algorithm is measured using M. W. Carter et al. (1996) datasets which is also known as Toronto Benchmark dataset that can be freely downloaded from: <http://www.asap.cs.nott.ac.uk/resources/data.shtml>. They introduced 12 real world case studies. However, there was some confusion in the literature due to some datasets appearing in different versions but under the same names as reported by R. Qu et al. (2009). These versions are classified as version *I*, *II* and *IIc* where the main differences between them are in terms of number of students, number of enrolments, number of examinations and conflict density R. Qu et al. (2009). Datasets from version *I* have been considered as shown in Table 2.

Note that, the conflict matrix density in the last column is calculated as the average number of other examinations that each examination conflicts with, divided by the total number of examinations (R. Qu et al., 2009). For example, a conflict matrix density of 0.5 or 50% indicates that, on average, each examination conflicts with half of the other examinations.

The main factor in the evaluation process is the solution quality, which is measured by calculating the violations of soft constraints in feasible solution. The penalty costs are used to compare between the proposed algorithms and other existing techniques found in the literatures. Other factors that influence of the proposed IWD algorithms include: (a) Number of IWD, which equals to the number of timetabling solutions, (b) The maximum number of iterations, which is predefined in the initial stage of search.

The penalty value of each solution is calculated by the total number of the soft constraints' violations in a feasible timetable that satisfied the hard constraints using Equation 1. The penalty values of the solutions obtained by the proposed IWD algorithms were compared with the most recent techniques using Carter datasets.

5.1. Experimental design

Experiments showing the effectiveness of Hybrid IWD algorithm. The study that examines the effect of varying LSR parameter on the performance of Hybrid IWD algorithm is conducted. Table 3 shows the details of parameter settings, which was adopted from the values of LSR that need to be evaluated. Note that, the number of solutions was fixed based on the settings that achieved best results (Drops = 25) as discussed in Bashar A. AlDeeb (2016).

The proposed hybrid IWD algorithm has been experimented in three scenarios due to the variation of Local Search Ratio (LSR) which denotes to the ratio of hybridizing the local search optimized with the hybrid IWD algorithm. LSR = 5%, LSR = 20% and LSR = 40% are selected as shown in Table 3. Each scenario runs

10 times on all Carter datasets (Total = 360 runs: 3 scenarios $\times$ 12 datasets $\times$ 10 solutions).

The results of the three scenarios are recorded in Table 4. The results include the best penalty value, the worst, the average of all penalty values and the standard deviation for each dataset and scenario. The total number of runs for the three proposed IWD algorithms was 1440 runs (Total = 1440 runs: 12 scenarios $\times$ 12 datasets $\times$ 10 solutions). The proposed intelligent water drops algorithms were experimented in Microsoft Visual C++ version 6.0 under Windows 7 on a Core i7 processor with 8 GB of RAM.

6. Comparative evaluation

The comparisons of outcomes are conducted with the approaches that used the same dataset in the literature to solve the UETP. Where the researchers compared the results of the proposed algorithm with Swarm based approaches, Hybrid approaches and other Metaheuristics approaches. Then the performance of the proposed IWD algorithms compared with the best-known results of algorithms that used the same database. It is worthy of mentioning that these results are the best proximity

Table 3
Hybrid Modified IWD algorithm Convergence Scenarios.

Scenario	Number of Iterations	Local Search Rate (LSR)	Number of Solutions
Scenario 1	1000	5%	25 solutions*
Scenario 2	1000	20%	25 solutions
Scenario 3	1000	40%	25 solutions

*Note: The number of solutions = 25 because it achieved the best result in Modified IWD algorithm.

Table 4
Comparison of the experimental results between Adapted IWD, Modified IWD and Hybrid IWD algorithm.

Dataset	Adapted IWD algorithm	Modified IWD algorithm	Hybrid IWD algorithm
CARS91I	7.82	5.48	5.41
CARF92I	6.51	4.59	4.59
EARF83I	47.7	35.79	33.64
HECS92I	13.48	10.35	10.15
KFUS93	20.54	14.35	13.49
LSEF91	16.8	11.4	10.44
RYES93	15.91	8.98	8.93
STAF83I	157.22	157.03	157.03
TRES92	11.17	8.69	8.34
UTAS92I	5.11	3.73	3.72
UTES92	32.49	25.06	24.81
YORF83I	46.57	38.11	36.65

Table 2
The cater dataset characteristics.

Institution (Abbreviation)	Timeslots	Exams	Students	Density
Ecole des Hautes Etudes Commerciales, Montréal (HECS92I)	18	81	2823	0.42
St.Andrew's Junior High School, Toronto (STAF83I)	13	139	611	0.14
York Mills Collegiate Institute, Toronto (YORF83I)	21	181	941	0.29
Faculty of Engineering, University of Toronto (UTES92)	10	184	2750	0.08
Earl Haig Collegiate Institute, Toronto (EARF83I)	24	190	1125	0.27
Trent University Peterborough, Ontario (TRES92)	23	261	4360	0.18
London School of Economics (LSEF91)	18	381	2726	0.06
King Fahd University, Dharan (KFUS93)	20	461	5349	0.06
Ryerson University, Toronto (RYES93)	23	481	11,483	40.07
Carleton University, Ottawa (CARF92I)	32	543	18,419	0.14
Faculty of Arts and Sciences, University of Toronto (UTAS92I)	35	622	21,266	0.13
Carleton University, Ottawa (CARS91I)	35	682	16,925	0.13

costs of the solutions recorded for each IWD algorithms are shown in Table 4. The best proximity cost obtained is highlighted in **bold**.

6.1. Results comparison with IWD versions

The proximity cost obtained by the Adapted IWD algorithm is the worst when compared with those obtained by the Modified IWD and Hybrid IWD algorithms on all Carter datasets (See Table 4). The Hybrid IWD can achieve the lowest proximity costs for all datasets, when compared with Adapted IWD and Modified IWD algorithms. Again, it showed that the hybridization of LSO after the solution construction phase in Modified IWD algorithm has significant effect on the convergence behavior of Modified IWD algorithm.

In addition, combination of Global best operator within the solution construction phase in Adapted IWD algorithm has also significant effect on the convergence behavior of Adapted IWD

algorithm. The Modified IWD algorithm achieved lower proximity cost than Adapted IWD for all the datasets. This showed that global best operator is effective for the enhancement of the results.

Fig. 1 visualize the comparative results as a bar chart that summarizes the proximity costs obtained by each proposed method for each dataset. It can be observed from the figure that Hybrid IWD algorithm obtained lowest proximity costs for all carter datasets, followed by Modified IWD algorithm and Adapted IWD algorithm. This shows that the Hybrid IWD algorithm performs better across the problem instance of the carter datasets.

6.2. Results comparison with other swarm based approaches

The swarm intelligence depends on the behavior of the self-organized systems' to develop metaheuristics that simulates such system's problem solving (Rozenberg et al., 2009). The characteristics of swarm intelligence inspired a number of researchers to

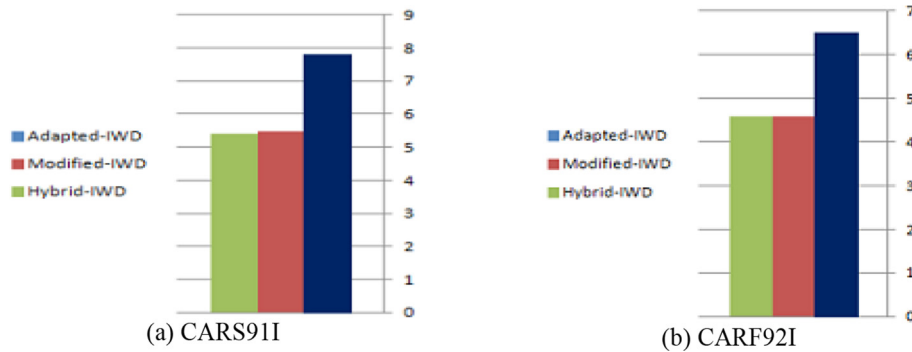


Fig. 1. Penalty Value of the three proposed methods for some datasets.

Table 5
Comparisons with Swarm based approaches.

Dataset	Adapted IWD	Modified IWD	Hybrid IWD	Eley (2007) (S1)	Turabieh and Abdullah (2011b) (S2)	Alzaqebah and Abdullah (2011) (S3)	Alinia Ahandani et al. (2012) (S4)	Bolaji et al. (2012) (S5)	Salwani Abdullah and Alzaqebah (2013) (S6)	Nasser R. Sabar et al. (2012) (S7)	Abed and Alicia (2013) (S8)	Bolaji, Khader, Al-Betar, and Awadallah (2015) (S9)
CARS91I	7.82	5.48	5.41	5.2	5.7	5.42	3.22	5.38	4.76	4.79	5.38	5.00
CARF92I	6.51	4.59	4.59	4.3	4.8	4.84	4.67	4.61	3.94	3.9	4.54	4.22
EARF83I	47.7	35.79	33.64	36.8	36.8	37.54	35.74	38.58	33.61	34.69	36.27	34.08
HECS92I	13.48	10.35	10.15	11.1	11.3	11.21	10.74	11.17	10.56	10.66	10.73	10.3
KFUS93	20.54	14.35	13.49	14.5	15	15.13	14.47	14.89	13.44	13	15.17	13.9
LSEF91	16.8	11.4	10.44	11.3	12.1	12.1	10.76	11.74	10.87	10	11.87	11.04
RYES93	15.91	8.98	8.93	9.8	10.2	–	9.95	9.8	8.81	–	8.6	9.18
STAF83I	157.2	157.03	157.03	157.3	157.2	157.5	157.1	157.2	157.09	157.04	158.16	157.04
TRES92	11.17	8.69	8.34	8.6	8.8	9.23	8.47	8.96	7.94	7.87	8.86	8.38
UTAS92I	5.11	3.73	3.72	3.5	3.8	3.94	3.52	3.65	3.27	3.1	3.59	3.40
UTES92	32.49	25.06	24.81	26.4	27.7	27.57	25.86	26.89	25.36	25.94	25.34	28.8
YORF83I	46.57	38.11	36.65	39.4	39.6	40.94	38.72	39.34	35.74	36.15	40.26	36.5

Table 6
Rank results achieved by the proposed IWDs against the Swarm based approaches.

Algorithm	CARS91I	CARF92I	EARF83I	HECS92I	KFUS93	LSEF91	RYES93	STAF83I	TRES92	UTAS92I	UTES92	YORF83I
Adapted IWD	12th	12th	12th	12th	12th	12th	10th	6th	12th	12th	12th	12th
Modified IWD	8th	5th	4th	2nd	4th	6th	4th	1st	6th	8th	2nd	4th
Hybrid IWD	7th	5th	1st	1st	2nd	2nd	3rd	1st	2nd	7th	1st	3rd
Best Results	5.41	4.59	33.64	10.15	13.49	10.44	8:93	157.03	8.34	3.72	24.81	36.65
Best result cited	3.22	3.9	34.2	10.4	13	10	8.6	157.1	7.87	3.1	25.3	36.15
Differences Ratio (%)	40.48	15	3.39	2.46	3.63	4.21	3.7	0.04	5.64	16.67	1.98	1.36

Table 7

Dataset	Adapted IWD	Modified IWD	Hybrid IWD	Merlot et al. (2003) (H1)	Qu and Burke (2005) (H2)	Salwani and Dorr (2007) (H3)	Qu and Burke (2009) (H4)	Qu et al. (2009a) (H5)	E. K. Burke et al. (2010) (H6)	Turabieh and Abdullah (2011b) (H7)	Turabieh and Abdullah (2011a) (H8)	Malek and Abdullah (2011) (H9)	Abed and Alicia (2013) (H10)	Mohammed Azmi Al-Betar et al. (2014) (H11)	Lei et al. (2014) (H12)	Burke, Qu, M. and Soghier (2013) (H13)	Alzaqebah and Abdullah (2015) (H14)	C. W. Fong, Asmuni, and McCollum (2015) (H15)	Bolaji et al. (2015) (H16)	Leite et al. (2018) (H17)
CARS911	7.82	5.48	5.41	5.1	5.3	5.2	5.16	5.11	4.6	4.8	4.81	5.33	5.38	4.99	6.4	5.19	4.38	4.79	5	4.311
CARF921	6.51	4.59	4.59	4.3	4.77	4.4	4.16	4.32	4	4.1	4.11	4.39	4.54	4.29	5.4	4.31	3.88	3.89	4.22	3.68
EARF931	4.7	35.79	33.64	35.1	38.39	34.9	35.86	35.56	32.8	34.92	36.1	35.17	36.27	34.42	39.8	35.79	33.34	33.43	34.07	32.48
HECS921	13.48	10.35	10.15	10.6	12.01	10.3	11.94	11.62	10	10.73	10.95	11.19	10.73	10.4	11.2	11.19	10.39	10.49	10.36	10.03
KFUS93	20.54	14.35	13.49	13.5	15.09	13.5	14.79	15.81	13	13	13.21	14.07	15.17	13.5	16	14.51	13.23	13.72	14.01	13.80
LSEF91	16.8	11.4	10.44	10.5	12.72	10.2	11.15	11.32	10	10.01	10.2	11.89	11.87	10.48	13.7	10.92	10.52	10.29	11.01	10.78
RYES93	15.91	8.98	8.93	-	-	-	-	-	-	-	-	-	8.6	8.79	12.1	-	8.92	-	9.28	-
STAT931	157.22	157.03	157.3	157.3	158.1	159.2	159	158.88	159.9	158.26	159.74	157.39	158.16	157.04	157.2	157.18	157.06	157.07	157.04	157.04
TRES92	11.17	8.69	8.34	8.4	8.74	8.4	8.6	8.52	7.9	7.88	8	9.41	8.86	8.16	9.5	8.49	7.89	7.86	8.38	8.34
UTAS921	5.11	3.73	3.72	3.5	3.32	3.6	3.59	3.21	3.2	3.2	3.32	3.89	3.59	3.43	4.3	3.44	3.13	3.1	3.4	3.4
UTES92	32.49	25.06	24.81	25.1	30.32	26	28.3	28	24.83	27	26.17	27.11	25.34	25.09	27.3	26.7	25.12	25.33	25.8	24.82
YORF831	46.57	38.11	36.65	37.4	40.24	36.2	41.81	40.71	37.28	36.22	36.23	40.76	40.26	35.86	40.5	39.47	35.49	36.12	36.95	34.45

employ such behavior in the algorithms for timetabling problems (Alzaqebah & Abdullah, 2015; Weng & Asmuni, 2013).

In recent years, several swarm-based approaches have been employed to solve the university examination timetabling problem, and some of these approaches were tested against [Carter et al. \(1996\)](#) datasets. In this section, we evaluate the proposed IWD algorithm by comparing the results with the results obtained by the previous swarm-based approaches that were tested using Carter dataset (See [Table 5](#)).

Table 5 represents the comparison between the proposed IWD algorithms and the previous swarm-based approaches. The symbol of (-) indicates that did not generate a feasible solution for this dataset. The value highlighted showed the best solution obtained for that problem instance. That table shows that the Adapted IWD algorithm has the ability of producing good quality results, which are within the range of the other existing approaches that used the same datasets. On the other hand, the third column represents the results obtained by the Modified IWD algorithm which were very good and competitive comparison when compared with those obtained by other swarm-based approaches. The Modified IWD algorithm achieved the best result in one instance of Carter dataset (*STAF83I*). Therefore, the Modified IWD achieved a competitive rank within the swarm-based approaches that are applies to the UETP as summarized in **Table 6**.

The comparison of the Hybrid IWD algorithm with other swarm-based approaches showed that there is superiority of the Hybrid IWD. As proof of this superiority, the fourth column of the table indicates that the Hybrid IWD obtained the best results in three instances (*HECS92I*, *STAF83I* and *UTES92*). Additionally, the improvement ratio for the Hybrid IWD algorithm was between 0.04% and 3.39%. Also, it achieved a good rank when compared to other swarm-based approaches (See [Table 6](#)).

6.3. Comparison with Hybrid approaches

In the last decade, many timetabling researchers have hybridized the local search based on population-based approaches. In this research, hybridization between a local search optimizer and the Modified IWD algorithm was done to achieve further improvement in the exploitation capability, and we called the proposed algorithm (Hybrid IWD). To evaluate the Hybrid IWD algorithm we make comparison between the results of the proposed Hybrid IWD and the results of the pervious approaches that were tested using Carter dataset.

Table 7 shows that the Hybrid IWD algorithm (fourth column) obtained two best results (STAF831 = 157.03 and UTES92 = 24.82) in comparison with the best-known results for the Hybrid approaches in the literature, and the percentage of the improvement was between (0.01% and 6.12%). It achieved the second rank in one dataset (See Table 8). Overall, the Adapted IWD and Modified IWD algorithms obtained good and competitive results.

6.4. Comparison with other metaheuristics approaches

Interestingly, the metaheuristic approaches have enlarged due to their ability to create solutions that are better than those created from the sequential heuristics alone (Schaefer, 1999). Usually in the university timetabling, an initial solution is generated using an appropriate heuristic. Then the enhancement is accomplished using these metaheuristics. In recent years, researchers have proposed and applied many metaheuristic approaches for solving UETP. Therefore, it is important to make comparisons between the proposed IWD algorithms and the metaheuristics approaches in the literature that tested for UETP using the Carter dataset (See Table 9).

Table 8

Rank results achieved by the proposed IWDs against the Hybrid techniques.

Algorithm	CARS91I	CARF92I	EARF83I	HECS92I	KFUS93	LSEF91	RYES93	STAF83I	TRES92	UTAS92I	UTES92	YORF83I
Adapted IWD	20th	20th	20th	20th	20th	20th	10th	10th	20th	20th	20th	20th
Modified IWD	18th	16th	13th	4th	13th	14th	6th	1st	15th	17th	3rd	12th
Hybrid IWD	17th	16th	3rd	2nd	4th	6th	5th	1st	7th	15th	1st	8th
Best Results	5.41	4.59	33.64	10.15	13.49	10.44	8.93	157.03	8.34	3.72	24.81	36.65
Best result cited	4.311	3.68	32.48	10	13	10	8.6	157.04	7.86	3.1	24.83	34.45
Differences Ratio (%)	19	15.46	2.56	1.5	3.77	4.4	3.84	0.01	6.1	20	6.12	3.26

Table 9

Comparisons with other metaheuristics approaches.

Dataset	Adapted IWD	Modified IWD	Hybrid IWD	Di Gaspero and Schaefer (2001) (Q1)	Di Gaspero (2002) (Q2)	Casey and Thompson (2003) (Q3)	Burke and Newall (2003) (Q4)	Burke et al. (2004) (Q5)	S. Abdullah, Ahmadi, Eley Burke, Dror, and McCollum (2007) (Q6)	N. Pillay and Banzhaf (2007) (Q7)	N. Pillay and Banzhaf (2010) (Q8)	Mohammed Azmi Al-Betar, Khader, and Thomas (2010) (Q9)	Tilahun (2019)
CARS91I	7.82	5.48	5.41	6.2	5.7	5.4	4.65	4.8	5.2	5.2	4.92	4.99	
CARF92I	6.51	4.59	4.59	5.2	–	4.4	4.1	4.2	4.4	4.3	4.22	4.29	5.91
EARF83I	47.7	35.79	33.64	45.7	39.4	34.8	37.05	35.4	34.9	36.8	35.87	34.42	
HECS92I	13.48	10.35	10.15	12.4	10.9	10.8	11.54	10.8	10.3	11.1	11.5	10.4	11.54
KFUS93	20.54	14.35	13.49	18	–	14.1	13.9	13.7	13.5	14.5	14.37	13.5	–
LSEF91	16.8	11.4	10.44	15.5	12.6	14.7	10.82	10.4	10.2	11.3	10.89	10.48	–
RYES93	15.91	8.98	8.93	–	–	–	–	8.9	8.7	9.8	9.3	8.79	–
STAF83I	157.22	157.03	157.03	160.8	157.4	–	168.73	159.1	159.2	157.3	157.81	157.04	–
TRES92	11.17	8.69	8.34	10	–	8.7	8.35	8.3	8.4	8.6	8.38	8.16	7.96
UTAS92I	5.11	3.73	3.72	4.2	4.1	–	3.2	3.4	3.6	3.5	3.35	3.43	3.57
UTES92	32.49	25.06	24.81	27.8	–	25.4	25.83	25.7	26	26.4	27.24	25.09	–
YORF83I	46.57	38.11	36.65	41	39.7	37.4	37.28	36.7	36.2	39.4	39.33	36.86	–

As shown in Table 9, the Hybrid IWD algorithm performed better than the previous metaheuristics algorithms. The Hybrid IWD obtained five best results ($EARF83I = 33.64$, $HECS92I = 10.15$, $KFUS93 = 13.49$, $STAF83I = 157.03$ and $UTES92 = 24.81$). Hybrid IWD algorithm achieved good ranks between the comparative methods (i.e., the 1st rank in five instances, the 2nd rank in one instance and the 3rd rank in two instances) (See Table 10). On the other hand, the Modified IWD algorithm achieved the 1st rank in one instance. Therefore, the proposed IWD algorithms outperformed many metaheuristics approaches reported in the literature.

6.5. Comparison with the Best-Known results

This section compares the performance of the proposed three IWD algorithms with the best-known results in the literature for the uncapacitated datasets as shown in Table 11. A statistical test was also carried out to show the significant differences between the compared approaches. The proposed IWD algorithms (Adapted IWD, Modified IWD and Hybrid IWD) were compared with 36 comparative approaches taken from the literature as shown in Tables 5, 7 and 9. Table 4 represents the results obtained by the proposed IWD algorithms on Carter et al. (1996). The results are found to be competitive in comparison with the results in the literatures (See Table 11). This shows that Adapted IWD algorithm has the ability of producing feasible solutions. However, the Friedman test (See Table 12) shows that the Adapted IWD algorithm needs some improvement to provide high quality results to be more competi-

tive with the results of the other approaches that are used to solve the university exam timetabling problem.

As shown in Table 11, the results obtained by the Modified IWD algorithm are very good and competitive in comparison with those obtained by the approaches in the literatures. The Modified IWD algorithm achieved the best results in one Carter dataset (i.e., STAF83I, PV = 157.03) and these results are better than those of the approaches in Tables 5, 7 and 9. Therefore, the Modified IWD achieved a competitive rank within the approaches applied for UETP (See Table 12).

The Hybrid IWD algorithm enhanced the quality of the solutions that are achieved by the Modified IWD algorithm. When this algorithm is compared with the swarm-based approaches and the hybrid approaches, it achieved the third and sixth ranking consecutively. Further, it achieved the first ranking in comparison with other metaheuristics (See Table 12). Perhaps, it is worthy of mentioning that the performance by the Hybrid IWD algorithm is more consistent across all 12 Carter datasets, where it achieved good rank. This is an indication that the Hybrid IWD algorithm is capable of producing high quality results across the Carter et al. (1996) datasets.

7. Conclusion and future work

The overall concern of this study was to investigate on how the proposed algorithms could improve the available search methodologies for the examination timetabling problems. The main goal

Table 10

Rank results achieved by the proposed IWDs against metaheuristics approaches.

Algorithm	CARS91I	CARF92I	EARF83I	HECS92I	KFUS93	LSEF91	RYES93	STAF83I	TRES92	UTAS92I	UTES92	YORF83I
Adapted IWD	12th	12th	14th	14th	13th	14th	8th	4th	13th	12th	13th	14th
Modified IWD	10th	9th	7th	3rd	8th	9th	5th	1st	9th	9th	2nd	8th
Hybrid IWD	9th	9th	1st	1st	1st	3rd	4th	1st	3rd	8th	1st	2nd
Best Results	5.41	4.59	33.64	10.15	13.49	10.44	8.93	157.03	8.34	3.72	24.81	36.65
Best result cited	4.65	4.1	34.42	10.3	13.5	10.2	8.7	157.04	7.96	3.2	25.09	36.2
Differences Ratio (%)	16.8	15	2.5	1.48	3.63	4.21	4.48	0.01	5.52	15.59	0.04	2.16

Table 11

Comparisons with Best Known results.

Dataset	Adapted IWD	Modified IWD	Hybrid IWD	Best Known	References
CARS911	7.82	5.48	5.41	3.22	Alinia Ahandani et al. (2012)
CARF921	6.51	4.59	4.59	3.68	Alzaqebah and Abdullah (2015)
EARF831	47.7	35.79	33.64	32.48	E. K. Burke et al. (2010)
HECS921	13.48	10.35	10.15	10	E. K. Burke et al. (2010)
KFUS93	20.54	14.35	13.49	13	(E. K. Burke et al., 2010; Nasser R. Sabar et al., 2012; Turabieh & Abdullah, 2011a)
LSEF91	16.8	11.4	10.44	10	(E. K. Burke et al., 2010; Nasser R. Sabar et al., 2012)
RYES93	15.91	8.98	8.93	7.3	Carter et al. (1996)
STAF831	157.22	157.03	157.03	157.04	(Al-Betar et al., 2010; Bolaji et al., 2015)
TRES92	11.17	8.69	8.34	7.86	Fong et al. (2015)
UTAS921	5.11	3.73	3.72	3.1	(Fong et al., 2015; Nasser R. Sabar et al., 2012)
UTES92	32.49	25.06	24.81	24.83	E. K. Burke et al. (2010)
YORF831	46.57	38.11	36.65	34.45	Alzaqebah and Abdullah (2015)

Table 12

Average ranking of the algorithms (Friedman Test) on the Carter datasets.

No.	Algorithm	Mean Rank	Ranking	No.	Algorithm	Mean Rank	Ranking	No.	Algorithm	Mean Rank	Ranking
1	<i>Adapted IWD</i>	11.64	12th	1	<i>Adapted IWD</i>	19	20th	1	<i>Adapted IWD</i>	13	13th
2	<i>Modified IWD</i>	5.55	6th	2	<i>Modified IWD</i>	11.68	13th	2	<i>Modified IWD</i>	6	5th
3	<i>Hybrid IWD</i>	3.09	3rd	3	<i>Hybrid IWD</i>	6.36	6th	3	<i>Hybrid IWD</i>	1.75	1st
4	S1	4.77	4th	4	H1	9.18	9th	4	O1	12	12th
5	S2	6.55	7th	5	H2	10.41	11th	5	O2	9.25	10th
6	S3	9.45	10th	6	H3	15.5	18th	6	O3	9.5	11th
7	S4	10.32	11th	7	H4	9.55	10th	7	O4	6.38	6th
8	S5	5.18	5th	8	H5	14.27	15th	8	O5	7.5	7th
9	S6	8.23	9th	9	H6	13.55	14th	9	O6	3.88	4th
10	S7	2.73	1st	10	H7	4.18	2nd	10	O7	2	2nd
11	S8	2.73	1st	11	H8	6	3rd	11	O8	8.25	9th
12	S9	7.77	8th	12	H9	8.55	8th	12	O9	8	8th
				13	H10	14.68	17th	13	O10	3.5	3rd
				14	H11	14.59	16th				
				15	H12	6.27	5th				
				16	H13	17.18	19th				
				17	H14	11.64	12th				
				18	H15	3.64	1st				
				19	H16	6	3rd				
				20	H17	7.77	7th				

behind the use of intelligent water drop algorithms is to determine how the IWD algorithm could be employed to improve the state-of-the-art solutions and thus provide the timetabling community with alternative approaches to solve the problem. The study is carried out by examining the proposed algorithm to improve the performance and effectiveness of the IWD algorithm for the Carter dataset.

The researchers initially modeled the examination timetabling problem in terms of the minimization objective function. Thereafter, the IWD algorithm was adapted to be workable for the UETP. Therefore, some operators have been adapted to maintain the feasibility (AlDeeb, 2016). To improve the process of assigning timeslot to next exam in the solutions construction phase using an idea that stemmed from the global-best concept of particle swarm optimization to improve the convergence rate. During the solutions construction phase, the global best solutions have intensively been used to share the best elements in the constructed solutions. The global best solution will attract the other constructed solutions to a better region in the search space and thus, will improve the quality of the solutions (AlDeeb, 2016). On the other side, the hybrid technique is also recommended for the timetabling domain to improve the exploitation capability. Therefore, the Hybrid IWD algorithm is proposed to improve the exploitation capability. By means of hybridizing Local Search Optimizer (LSO) after the solutions construction process to improve the constructed solution locally controlled by Local Search Rate (LSR). In the LSO three successful neighborhood structure were used (Move, Swap, Kempe chain). Note that, the LSR is the rate of using the LSO. The perfor-

mance of the Hybrid IWD is evaluated using the Carter dataset on three convergent scenarios of varying LSR (5%, 20% and 40%). It is found that increasing the value of LSR leads to better results. A comparison between the Adapted IWD, the Modified IWD and the Hybrid IWD was conducted. In all Carter datasets, the Hybrid IWD technique achieved the best results.

A comprehensive experimental analysis against on 36 published approaches, these approaches include 9 swarm-based approaches, 10 metaheuristics methods and 17 Hybrid approaches which used the Carter datasets, showed that the proposed Hybrid IWD algorithms achieved a good rank among the state-of-the-art approaches. It is noteworthy that the Hybrid IWD algorithm achieved the first rank by using the Friedman test. Further, the algorithm attained three best results in comparison with the swarm intelligent approaches (the percentage of the improvement was between 0.04% and 2.46% and the Hybrid IWD achieved the second rank) and achieved one best results (with the improvement of 0.01% and the second rank for the Hybrid IWD) in comparison to the other metaheuristics approaches in the literature. These results indicate that the efficiency of the intelligent system using the Hybrid IWD algorithm was better than that of the manual method.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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